Non-Parametric Least Squares

Based on Chapter 13 of *High Dimensional Statistics* by Martin Wainwright

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Section 1

Introduction

Non-Parametric Least Squares

- **MMSE Estimation**:

$$\bar{\mathcal{L}}_f = \mathbb{E}_{X,Y}[(Y - f(X))^2] \implies f^\star(x) = \mathbb{E}[Y|X = x]$$

- In practice we are given a collection of samples $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, which can be used to compute an empirical analog of the MSE:

$$\widehat{\mathcal{L}}_f = \frac{1}{n}\sum_{i=1}^{n}\big(y_i - f(x_i)\big)^2$$

- **Non-Parametric Least Squares**: Minimizing this criterion over some suitably controlled function class.

Different Measures of Quality

- **Excess Risk**:

$$||f - f^\star||_{L^2(\mathbb{P})} = \mathbb{E}_{X\sim\mathbb{P}}\Big[\big(f(X) - f^\star(X)\big)^2\Big]$$

where $\mathbb{P}$ denotes the distribution over covariates.

- **This Chapter**: Let $\{\mathbf{x}_i\}_{i=1}^n$ be the set of fixed covariates and $\mathbb{P}_n$ be their Empirical measure. Define:

$$||f - f^\star||_{L^2(\mathbb{P}_n)} = \left[\frac{1}{n}\sum_{i=1}^n \big(f(\mathbf{x}_i) - f^\star(\mathbf{x}_i)\big)^2\right]^{1/2}.$$

In this talks, we will use this measure and denote it as $||f - f^\star||_n$.

Fixed or Stochastic Covariates?

Note

- We will view the samples $\{\mathbf{x}_i\}_{i=1}^n$ as being fixed, a set-up known as *regression with a fixed design*.
- Results from Chapter 14 to follow can be used to translate these bounds into equivalent results in the population $L_2(\mathbb{P})$-norm.

Estimation via Least Squares

Given a fixed collection $\{\mathbf{x}_i\}_{i=1}^n$, model the responses as

$$y_i = f^\star(\mathbf{x}_i) + \nu_i, \quad \text{for } i = 1, 2, \ldots, n.$$

where $\nu_i = \sigma w_i$ in which $w_i \sim \mathcal{N}(0,1)$. The least squares estimate is given by the function

$$\widehat{f} \in \arg\min_{f \in \mathcal{F}} \left\{ \frac{1}{n} \sum_{i=1}^n \big(y_i - f(x_i)\big)^2 \right\}.$$

- When $\nu_i \sim \mathcal{N}(0, \sigma^2)$, the LS estimate is equivalent to the constrained maximum likelihood.
- When $\mathcal{F}$ is an RKHS, it can also be convenient to use regularized estimators of the form:

$$\widehat{f} \in \arg\min_{f \in \mathcal{F}} \left\{ \frac{1}{n} \sum_{i=1}^n \big(y_i - f(x_i)\big)^2 + \lambda_n ||f||_{\mathcal{F}}^2 \right\}.$$

Section 2

Examples

Parametric: Linear Regression

- For a given vector $\boldsymbol{\theta} \in \mathbb{R}^d$, define $f_\theta(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle$ and consider the function class $\mathcal{F}_C = \{f_\theta : \mathbb{R}^d \to \mathbb{R} | \theta \in C\}$ for a compact C.
- The least squares estimate:

$$\widehat{\theta} \in \arg\min_{\theta \in C} \left\{ \frac{1}{n} ||y - \mathbf{X}\theta||_2^2 \right\}$$

where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the design matrix.

- The constrained form of Ridge Regression:

$$C = \left\{ \theta \in \mathbb{R}^d \;\middle|\; ||\theta||_2^2 \leq R \right\}.$$

- The constrained form of LASSO:

$$C = \left\{ \theta \in \mathbb{R}^d \;\middle|\; ||\theta||_1 \leq R \right\}.$$

Non-Parametric: Cubic Smoothing Spline

Consider the class of twice cont. differentiable functions $f\colon [0,1] \to \mathbb{R}$ and for a given radius R, define:

$$\mathcal{F}(R) = \left\{ f\colon [0,1] \to \mathbb{R} \;\middle|\; \int_0^1 \left(f''(x)\right)^2 dx \leq R \right\}.$$

The integral can be understood as a Hilbert norm bound in the second-order Sobolev space.

Section 3

Critical Radius: Bounding the Prediction Error

Critical Radius: Bounding the Prediction Error

- Intuitively, the difficulty of estimating the function $f^\star$ should depend on the complexity of the function class $\mathcal{F}$ in which it lies.
- **a localized form of Gaussian complexity:** it measures the complexity of the function class $\mathcal{F}$, locally in a neighborhood around the true regression function $f^\star$.

Local Gaussian Complexity

Define $\mathcal{F}^\star = \mathcal{F} - \{f^\star\} = \{f - f^\star | f \in \mathcal{F}\}$. For a given radius $\delta > 0$, *the local Gaussian complexity* around $f^\star$ at scale δ is given by:

$$\mathcal{G}_n(\delta; \mathcal{F}^\star) = \mathbb{E}_w \left[\sup_{g \in \mathcal{F}^\star, \, ||g||_n \leq \delta} \left| \frac{1}{n} \sum_{i=1}^{n} w_i \, g(\mathbf{x}_i) \right| \right]$$

Critical Inequality

Critical Inequality

A central object in our analysis is the set of positive scalars δ that satisfy the *critical inequality*:

$$\frac{\delta}{2\sigma} \geq \frac{\mathcal{G}_n(\delta; \mathcal{F}^\star)}{\delta}$$

- We refer to any $\delta_n > 0$ satisfying the critical inequality as being valid.
- We use $\delta_n^\star$ to denote the smallest positive radius for which the critical inequality holds.

Some Intuition

- Note that $\frac{1}{2n}\sum_{i=1}^{n}\left(y_i - \widehat{f}(\mathbf{x}_i)\right)^2 \leq \frac{1}{2n}\sum_{i=1}^{n}\left(y_i - f^\star(\mathbf{x}_i)\right)^2$.
- After some computations: $\frac{1}{2}||\widehat{f} - f^\star||_n^2 \leq \frac{\sigma}{n}\sum_{i=1}^{n} w_i\left(\widehat{f}(\mathbf{x}_i) - f^\star(\mathbf{x}_i)\right)$.
- Note that $\widehat{f} - f^\star \in \mathcal{F}^\star$. Reasoning hueristically, this observation suggests that

$$\frac{\delta^2}{2} \leq \sigma\mathcal{G}_n(\delta;\mathcal{F}^\star) \text{ or equivalently } \frac{\delta}{2\sigma} \leq \frac{\mathcal{G}_n(\delta;\mathcal{F}^\star)}{\delta}.$$

- We will present a rigorous result later in this talk.

The Main Result

Star-Shaped Classes

a function class $\mathcal{F}$ is star-shaped if for any $\alpha \in [0, 1]$ we have

$$f \in \mathcal{F} \implies \alpha f \in \mathcal{F}.$$

Theorem

Suppose that the shifted function class $\mathcal{F}^\star$ is star-shaped, and let δ_n be any solution to the critical inequality. Then for any $t \geq \delta_n$, the nonparametric least-squares estimate $\widehat{f}_n$ satisfies the bound

$$\mathbb{P}\left[||\widehat{f}_n - f^\star||_n^2 \geq 16t\delta\right] \leq \exp\left(\frac{-nt\delta}{2\sigma^2}\right) \tag{1}$$

Properties of Star-Shaped Classes

- The star shaped condition on $\mathcal{F}^\star$ is needed in various parts of the proof, including ensuring a valid radii δ_n exists.
- $\mathcal{F}^\star$ Star shaped $=$ $\mathcal{F}$ is star shaped around $f^\star$.

Property

If $\mathcal{F}$ is convex, it is star-shaped around any $f^\star$.

Proof

Let $\widetilde{f} = f - f^\star \in \mathcal{F}^\star$ be an arbitrary point in $\mathcal{F}^\star$ with $f, f^\star \in \mathcal{F}$.
For any $\alpha \in [0, 1]$, by convexity we have $g = \alpha f + (1 - \alpha) f^\star \in \mathcal{F}$.
Hence $\alpha \widetilde{f} = \alpha(f - f^\star) = g - f^\star \in \mathcal{F}^\star$, proving $\mathcal{F}^\star$ is star-shaped.

Conversely

Similarly if $\mathcal{F}$ is not convex, there must exists a $f^\star$ such that $\mathcal{F}$ is not star shaped around $f^\star$.

Properties of Star-Shaped Classes

- If the star-shaped condition fails to hold, then the main Theorem can instead by applied with δ_n defined in terms of the star hull (we will see next session.)

$$\mathrm{star}(\mathcal{F}^\star) = \{\alpha(f - f^\star) | f \in \mathcal{F}, \alpha \in [0,1]\}.$$

Existence of Critical Radius

For any star shaped class $\mathcal{F}^\star$, the function $\delta \to \frac{\mathcal{G}_n(\delta, \mathcal{F}^\star)}{\delta}$ is non-increasing on $(0, \infty)$. Consequently, for any $c > 0$, the inequality

$$\frac{\mathcal{G}_n(\delta, \mathcal{F}^\star)}{\delta} \leq c\delta$$

has a smallest positive solution.

Proof of the Existence of Critical Radius

Given $0 < \delta \leq t$, we should show that $\frac{\delta}{t}\mathcal{G}_n(t) \leq \mathcal{G}_n(\delta)$.
Given any $h \in \mathcal{F}^\star$ with $||h||_n \leq t$, define $\widetilde{h} = \frac{\delta}{t}h \in \mathcal{F}^\star$ (by star-shaped assumption) and write

$$\frac{1}{n}\Big\{\frac{\delta}{t}\sum_{i=1}^{n} w_i h(\mathbf{x}_i)\Big\} = \frac{1}{n}\Big\{\sum_{i=1}^{n} w_i \widetilde{h}(\mathbf{x}_i)\Big\}.$$

By construction, $||\widetilde{h}||_n \leq \delta$. Consequently:

- The RHS is at most $\mathcal{G}_n(\delta)$ in expectation.
- Taking supremum over the set $\mathcal{F} \cap \{||h||_n \leq t\}$ following by expectation, yields $\frac{\delta}{t}\mathcal{G}_n(t)$ on the left hand side.

This concludes the proof. $\square$

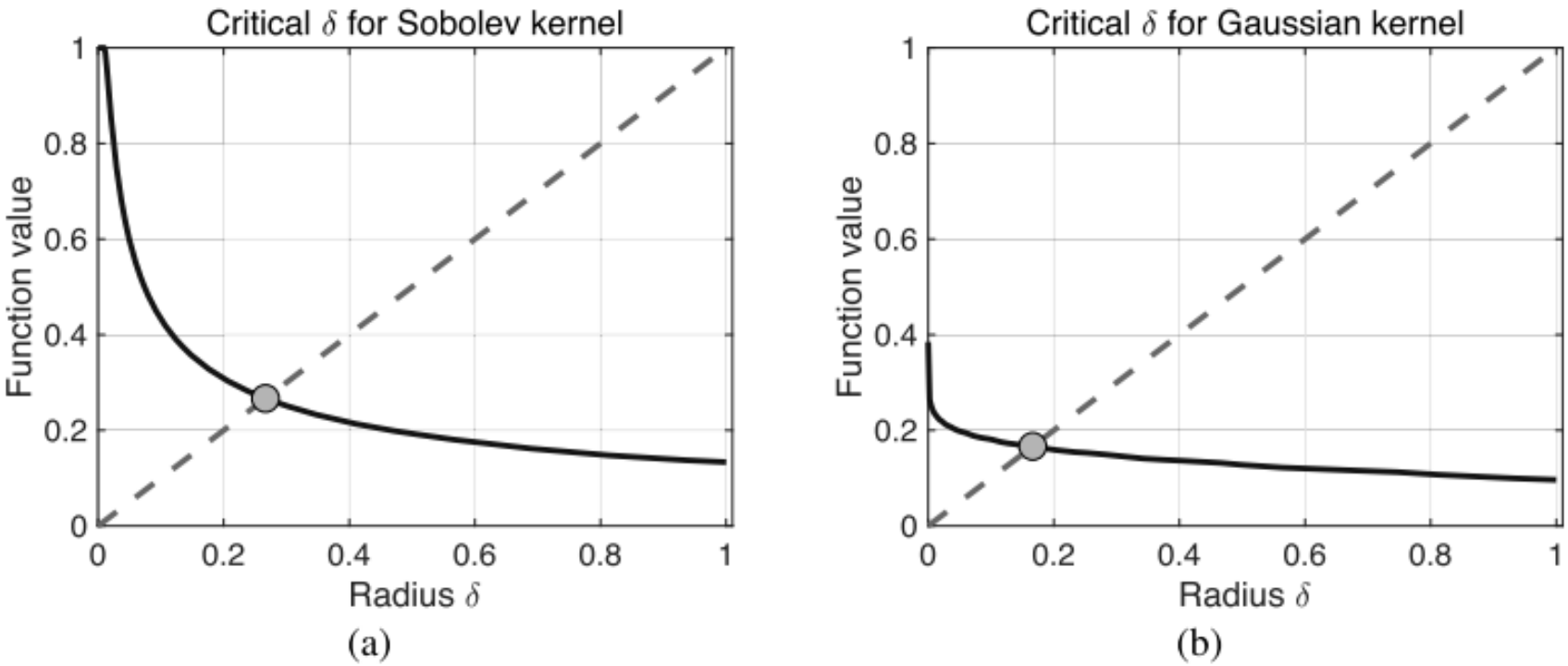

Figure 13.2 Illustration of the critical radius for sample size $n = 100$ and two different function classes. (a) A first-order Sobolev space. (b) A Gaussian kernel class. In both cases, the function $\delta \mapsto \frac{\mathcal{G}_n(\delta;\mathscr{F})}{\delta}$, plotted as a solid line, is non-increasing, as guaranteed by Lemma 13.6. The critical radius δ_n^*, marked by a gray dot, is determined by finding its intersection with the line of slope $1/(2\sigma)$ with $\sigma = 1$, plotted as the dashed line. The set of all valid δ_n consists of the interval $[\delta_n^*, \infty)$.

Proof of the main theorem

Theorem

Suppose that the shifted function class $\mathcal{F}^\star$ is star-shaped, and let δ be any solution to the critical inequality. Then for any $t \geq \delta$, the nonparametric least-squares estimate $\widehat{f}_n$ satisfies the bound

$$\mathbb{P}\left[||\widehat{f}_n - f^\star||_n^2 \geq 16t\delta\right] \leq \exp\left(\frac{-nt\delta}{2\sigma^2}\right)$$

- Denote $\widehat{\Delta} = \widehat{f} - f^\star \in \mathcal{F}^\star$, we have $\frac{1}{2}||\widehat{\Delta}||_n^2 \leq \frac{\sigma}{n}\sum_{i=1}^n w_i \widehat{\Delta}(\mathbf{x}_i)$. We need to control the stochastic component of the right-hand side.

Auxillary Lemma

Let $\mathcal{F}^\star$ be a star-shaped class and δ satisfy the critical inequality. Define the bad event $A(u)$ as

$$A(u) = \left\{ \exists g \in \mathcal{F}^\star, ||g||_n \geq u \ : \ \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2u||g||_n \right\}.$$

For a given $u \geq \delta$, we have $\mathbb{P}[A(u)] \leq \exp\left(\frac{-nu^2}{2\sigma^2}\right)$.

Now consider two cases:

- **Case 1:** $||\widehat{\Delta}||_n < \sqrt{t\delta}$ which implies $||\widehat{\Delta}||_n^2 \leq t\delta$ trivially.
- **Case 2:** $||\widehat{\Delta}||_n \geq \sqrt{t\delta}$. We condition on $A^c(\sqrt{t\delta_n})$. Set $u = \sqrt{t\,\delta}$, so that we have $\mathbb{P}[A^c(\sqrt{t\,\delta})] \geq 1 - \exp(\frac{-nt\delta}{2\sigma^2})$. Hence:

$$||\widehat{\Delta}||_n^2 \leq 2\left|\frac{\sigma}{n}\sum_{i=1}^n w_i \widehat{\Delta}(\mathbf{x}_i)\right| \leq 4||\widehat{\Delta}||_n\sqrt{t\delta} \implies ||\Delta||_n^2 \leq 16t\delta. \quad \square$$

Proof of the Auxiliary Lemma

Recall $A(u) = \left\{ \exists g \in \mathcal{F}^\star, ||g||_n \geq u \ : \ \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2u||g||_n \right\}$.

Step 1: reduce to controlling a sup over a subset with $||\widetilde{g}||_n \leq u$.

- Suppose that $A(u)$ is true, hence there exists $g \in \mathcal{F}^\star$ with $||g||_n \geq u$ such that

$$\left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2||g||_n u.$$

- Define $\widetilde{g} = \frac{u}{||g||_n} g \in \mathcal{F}^\star$, we observe that $||\widetilde{g}||_n = u$. We have

$$\left| \frac{\sigma}{n} \sum_{i=1}^n w_i \widetilde{g}(\mathbf{x}_i) \right| = \frac{u}{||g||_n} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i g(\mathbf{x}_i) \right| \geq 2u^2.$$

- We thus conclude that:

$$\mathbb{P}[A(u)] \leq \mathbb{P}[Z_n(u) \geq 2u^2], \ Z_n(u) := \sup_{\widetilde{g} \in \mathcal{F}^\star, \ ||\widetilde{g}||_n \leq u} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i \widetilde{g}(\mathbf{x}_i) \right|.$$

Proof of the Auxiliary Lemma

$\mathbb{P}[A(u)] \leq \mathbb{P}[Z_n(u) \geq 2u^2]$, $Z_n(u) := \sup_{\widetilde{g} \in \mathcal{F}^\star, \quad ||\widetilde{g}||_n \leq u} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i \widetilde{g}(\mathbf{x}_i) \right|$.

Step 2: Concentration of suprema.

- Recall that $w_i \sim \mathcal{N}(0,1)$, hence $\frac{\sigma}{n} \sum_{i=1}^n w_i \widetilde{g}(\mathbf{x}_i) \sim \mathcal{N}\left(0, \frac{\sigma^2}{n} ||\widetilde{g}||_n^2\right)$.
- $Z_n(u) = h(w_1, \ldots, w_n) := \sup_{\widetilde{g} \in \mathcal{F}^\star, \quad ||\widetilde{g}||_n \leq u} \left| \frac{\sigma}{n} \sum_{i=1}^n w_i \widetilde{g}(\mathbf{x}_i) \right|$.
- The Lipschitz constant of h is at most $\frac{\sigma u}{\sqrt{n}}$.
- We know that for $Z_i \sim \mathcal{N}(0, \sigma^2)$ and h a L-Lipschitz function:

$$\mathbb{P}[h(Z) - \mathbb{E}h(Z) \geq t] \leq e^{-\frac{s^2}{2L^2\sigma^2}}.$$

- Let $s = u^2$, we obtain $\mathbb{P}(Z_n(u) \geq \mathbb{E}[Z_n(u)] + u^2) \leq \exp\left(\frac{-nu^2}{2\sigma^2}\right)$.

Proof of the Auxiliary Lemma

Step 3: Bounding the expectation.

- Note that $\mathbb{E}[Z_n(u)] = \sigma\, \mathcal{G}_n(u)$.
- We know that $\nu \to \frac{\mathcal{G}_n(\nu)}{\nu}$ is non-increasing. We have also assumed that $u \geq \delta$.

$$\sigma \frac{\mathcal{G}_n(u)}{u} \leq \sigma \frac{\mathcal{G}_n(\delta)}{\delta} \leq \frac{\delta}{2} \leq \delta \implies \mathbb{E}[Z_n(u)] \leq u\delta \leq u^2.$$

Comibining our results, we obtain

$$\mathbb{P}[A(u)] \leq \mathbb{P}[Z_n(u) \geq 2u^2] \leq \mathbb{P}(Z_n(u) \geq \mathbb{E}[Z_n(u)] + u^2) \leq \exp\left(\frac{-nu^2}{2\sigma^2}\right),$$

Which concludes the proof. $\square$

Section 4

How to Compute the Critical Radius?

Bounds via Metric Entropy

For any star-shaped function class $\mathcal{F}^\star$, define:

- $B(\delta\,;\mathcal{F}^\star) = \{h \in \mathcal{F}^\star \mid ||h||_n \leq \delta\}$
- $\mathcal{N}\left(t; B_n(\delta; \mathcal{F}^\star)\right)$ be the t-convering number of $B_n(\delta; \mathcal{F})$ in norm $||.||_n$.

Critical Inequality via Metric Entropy

Any $\delta \in (0, \infty]$ satisfying

$$\frac{16}{\sqrt{n}} \int_{\frac{\delta^2}{4\sigma}}^{\delta} \sqrt{\log \mathcal{N}\left(t\,; B(\delta; \mathcal{F}^\star)\right)}\, dt \leq \frac{\delta^2}{4\sigma},$$

satisfies the critical inequality.

Bounds via Metric Entropy: Proof

- Construct a $\frac{\delta^2}{4\sigma}$-covering of the set $B(\delta;\mathcal{F})$, say $\{g^1,\ldots,g^M\}$. For any function $g \in B(\delta;\mathcal{F})$, there is a $j \in [M]$ such that $||g^j - g||_n \leq \frac{\delta^2}{4\sigma}$.
- We have

$$\begin{aligned}
\left|\frac{1}{n}\sum_{i=1}^n w_i\, g(\mathbf{x}_i)\right| &\leq \left|\frac{1}{n}\sum_{i=1}^n w_i\, g^j(\mathbf{x}_i)\right| + \left|\frac{1}{n}\sum_{i=1}^n w_i\left(g(\mathbf{x}_i) - g^j(\mathbf{x}_i)\right)\right| \\
&\leq \max_{j=1,\ldots,M}\left|\frac{1}{n}\sum_{i=1}^n w_i\, g^j(\mathbf{x}_i)\right| + \sqrt{\frac{\sum_{i=1}^n {w_i}^2}{n}}\sqrt{\frac{\sum_{i=1}^n (g(\mathbf{x}_i) - g^k(\mathbf{x}_i))^2}{n}} \\
&\leq \max_{j=1,\ldots,M}\left|\frac{1}{n}\sum_{i=1}^n w_i\, g^j(\mathbf{x}_i)\right| + \frac{\delta^2}{4\sigma}\sqrt{\frac{\sum_{i=1}^n {w_i}^2}{n}}.
\end{aligned}$$

Hence

$$\mathcal{G}_n(\delta) \leq \mathbb{E}\left[\max_{j=1,\ldots,M}\left|\frac{1}{n}\sum_{i=1}^n w_i\, g^j(\mathbf{x}_i)\right|\right] + \frac{\delta^2}{4\sigma}.$$

Bounds via Metric Entropy: Proof

- To upper bound the first term, we use Dudley's Integral. Define $Z(g^j) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} w_i g^j(\mathbf{x}_i)$ for $j = 1, \ldots, M$.
- The induced metric: $\rho_Z^2(g^j, g^k) := \mathrm{Var}\big(Z(g^j) - Z(g^k)\big) = ||g^j - gk||_n^2$
- Since $||g||_n \leq \delta$ for all $g \in B_n(\delta, \mathcal{F}^*)$, the coarsest resolution of the chaining can be set to δ. We can terminate it at $\frac{\delta^2}{4\sigma}$ since any member of our finite set can be reconstructed at this resolution.

$$\mathbb{E}\left[\max_{j=1,\ldots,M}\left|\frac{1}{n}\sum_{i=1}^{n} w_i g^j(\mathbf{x}_i)\right|\right] = \mathbb{E}\left[\max_{j=1,\ldots,M}\frac{\left|Z(g^j)\right|}{\sqrt{n}}\right]$$
$$\leq \frac{16}{\sqrt{n}}\int_{\frac{\delta^2}{4\sigma}}^{\delta}\sqrt{\log N_n(t; B_n(\delta; \mathcal{F}))}dt$$

Hence $\mathcal{G}_n(\delta) \leq \frac{\delta^2}{4\sigma} + \frac{16}{\sqrt{n}}\int_{\frac{\delta^2}{4\sigma}}^{\delta}\sqrt{\log N_n(t; B_n(\delta; \mathcal{F}))}dt$. □

Section 5

Examples

Bounds for Linear Regression

- Consider $y_i = \langle \boldsymbol{\theta}^\star, \mathbf{x}_i \rangle + w_i$ with $\boldsymbol{\theta}^\star \in \mathbb{R}^d$.
- Hypothesis class: $\mathcal{F} = \{ f_{\boldsymbol{\theta}}(.) = \langle \boldsymbol{\theta}, . \rangle \mid \boldsymbol{\theta} \in \mathbb{R}^d \}$.
- Let $\mathbf{X} \in \mathbb{R}^{d \times n}$ be the design matrix with rank r.
- $\mathcal{F} = \mathcal{F}^\star$ and convex. Thus star-shaped around any point.

Section 6

Oracle Inequalities

Section 7

Regularized Estimators